Michael C.

Senior Java Backend Engineer · Payments & Wallets · Distributed Systems

jcui1@my.hpu.edu · Portfolio: michael.opc-x.org · LinkedIn: linkedin.com/in/michael-java-backend · Open to remote roles

SUMMARY

Senior Java backend engineer with 13 years of experience, from Alibaba and DiDi (China's Uber) to crypto and licensed payments.

For the past 4 years I've owned wallet systems where every bug costs real money: a non-custodial multi-chain wallet at HTX/Huobi (~12M users) and a MAS-licensed e-wallet at dtcpay in Singapore.

I take a module from design to production on my own and work well in async, remote teams across time zones.

SKILLS

Java / Spring Boot	thread pools, JVM OOM troubleshooting, transaction pitfalls
MySQL	index design, cursor pagination, conditional updates + row locks
Redis	caching, distributed locks, Bloom filters
Kafka	reliable delivery, idempotent consumers
Payments	idempotency, state machines, channel integration, reconciliation
Blockchain	wallet creation, transaction broadcast, block scanning, reorg handling

EXPERIENCE

dtcpay — Senior Java Backend Engineer — E-Wallet Owner — Apr 2024 – Present · Singapore

MAS-licensed payment institution (fiat + stablecoin). Owned the core flow of the consumer e-wallet app: top-up → withdrawal → daily reconciliation; every balance change is booked by the wallet service.

- Concurrency safety (no overdraft): a single conditional UPDATE checks and deducts the balance under a row lock, so concurrent debits queue up and never go negative.
- Idempotent APIs: one-time Redis tokens block duplicate submits; a unique index on the business order ID is the final guard.
- Lost-callback recovery: verify signature and amount on channel callbacks; scheduled jobs query the channel to recover missed notifications.
- Idempotent payouts: freeze funds before payout; on channel timeout only query, never resend; deduct on success, unfreeze on failure.
- Eventual consistency: debit and pending notification are written in one DB transaction, then delivered via Kafka with retries (local message table).

HTX (formerly Huobi — crypto exchange) — Senior Java Backend Engineer — Blockchain Wallet Owner — May 2022 – Apr 2024

Huobi Wallet: a non-custodial multi-chain wallet with ~12M users (private keys stay on the user's phone). Owned the core flow: wallet creation → on-chain transfer → deposit detection → deposit notification.

- Chain reorgs: detect broken parentHash links, roll back and rescan; credit deposits only after enough confirmations.
- OOM fix: the block scanner crashed while catching up; root-caused an unbounded queue holding blocks in memory and switched to a bounded queue with back-pressure.
- Reliable delivery + idempotent consumption with Kafka: persist notifications before sending; each deposit is pushed exactly once.
- Address filtering with a Bloom filter: discard non-user addresses in each block before hitting the database.
- Scanner HA with a Redis distributed lock: one active scanner, automatic failover, resume from the saved cursor.

DiDi (China's Uber) — Senior Java Backend Engineer Oct 2018 – May 2022 · Hangzhou, China

Ride-hailing platform. Tech lead of the company-wide search service.

- Built the search service from 0 to 1 on Elasticsearch; weekly traffic grew from 60K to 100K.
- Built SQL lineage parsing that automatically extracts upstream/downstream table dependencies, exposed as a query service.
- Provided unified query APIs consumed by systems across the company.

Alibaba — Java Backend Engineer — Distributed Job Scheduling Oct 2016 – Oct 2018 · Hangzhou, China

Alibaba Group. Core developer of the scheduling system behind Alibaba Cloud DataWorks: master election, job sharding and timed triggers.

- Master HA: nodes compete for a database row lock (SELECT … FOR UPDATE); when the master dies the lock is released and another node takes over automatically.
- Job sharding: machines map to a fixed set of logical nodes, so scaling in/out only changes the mapping (similar to consistent hashing).
- Quartz-based scheduling of 1M+ hourly/daily/weekly/monthly jobs; average job development time dropped from 36h to 17.45h after launch.

Ourpalm (mobile games) — Java Backend Engineer Mar 2013 – Oct 2016 · Beijing, China

Mobile game company listed on the Shenzhen ChiNext board. Built the shared server side used by multiple games: log collection SDK and operations admin.

- Designed and built a server-side game log collection SDK adopted by multiple mobile games.
- Developed and maintained the Java operations admin system and its scheduled jobs.

EDUCATION

Hawaii Pacific University — M.S. Artificial Intelligence · GPA 3.88 / 4.0 · With Distinction Jan 2025 – Jul 2026

University of Electronic Science and Technology of China — B.Eng. Computer Science and Technology Sep 2008 – Jul 2012